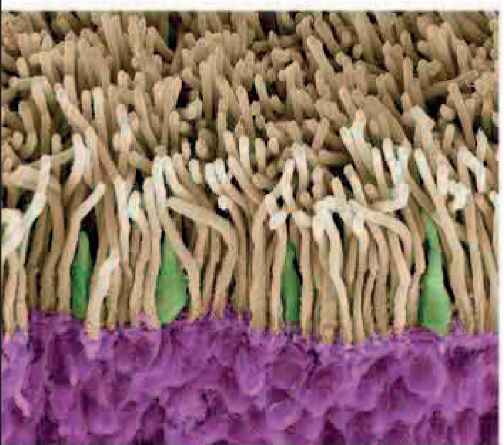
“ THOUGH **ORGANISMS**… CLAIM OUR PRIMARY INTEREST… WE CANNOT SEPARATE THEM FROM THEIR SPECIAL ENVIRONMENTS, WITH WHICH THEY **FORM ONE PHYSICAL SYSTEM.** ”

Arthur Tansley, from *Ecology Journal*, 1935

At the impressionable stage immediately after hatching, a brood of goslings assumed Konrad Lorenz to be their surrogate mother and followed him everywhere.



Retina rods and cones
Cells on the retina contain pigments for light-absorption; rods (sepia) have one kind, but cones (green) collectively have three kinds of pigment for detecting different colors.

nonliving environment that surrounded them.

Rudolf Schoenheimer (1898–1941) was a German biochemist studying **metabolism**—the complex pattern of chemical reactions in the body. In order to understand these patterns, he found a way of tagging, or marking, substances in the body with detectable isotopes (variants) of elements. By tracing the pathways of the isotopes, he was able to work out the sequences of chemical reactions taking place. **Isotopic labeling** would become the standard method for studying the biochemistry of metabolism.

Danish ophthalmologist **Gustav Østerberg** published a study on the cellular makeup of the retina at the back of the eye. In the 19th century, German anatomist Max Schultze had identified the layers of the retina and drawn detailed illustrations of structures later called rods and cones (see 1866). Østerberg recorded the **first accurate count of the rods and cones**. It was later shown that cones have high sensitivity in low light intensity, but cannot detect color. Color-sensitive cones only work in high light intensity and are concentrated in an area of the retina called the fovea, which collects light from a point of focus at the center of the field of vision to form an image.

7 MILLION CONES IN THE **HUMAN EYE**

130 MILLION RODS IN THE **HUMAN EYE**

IN 1936, DUTCH BIOLOGIST NIKOLAAS TINBERGEN (1907–88) met Austrian biologist **Konrad Lorenz** (1903–89) at a symposium, and the two men spent many months discussing aspects of **animal behavior**. Their collaboration marked the **foundation of ethology**—the modern science of animal behavior. They distinguished between innate behavior that was inherited and learned behavior that became modified through experience. Lorenz famously demonstrated how **goslings** can become attached (imprinted) to humans just after hatching; Tinbergen studied the innate **courtship behavior** of the **stickleback fish**.

The world’s **first practical helicopter**—the Focke-Wulf Fw61—took its maiden flight in June. Built by German aviator **Heinrich Focke** (1890–1979), it flew using twin rotary blades extending to the left and right of the fuselage. Its first flight lasted just 28 seconds—but it was far easier to control than previous versions.

In July, Hungarian biologist **Hans Selye** (1907–82) was the first person to describe the **scientific basis for physiological stress**. In the 19th century, French physiologist Claude Bernard had proposed that the living body maintained a steady state by processes of internal regulation. By the early 1900s it was understood how aspects of the nervous system could make the body respond to changes in circumstances—for example, initiating the “fight or flight” response to danger. Selye further explained how hormonal changes were associated with stress, and that these changes could affect the function of the body’s immune system.

In September, the last surviving **thylacine** died in Hobart Zoo in Tasmania, after being exposed to extreme weather conditions when a keeper inadvertently left it out of its shelter one night. Commonly known as the Tasmanian wolf or tiger, the thylacine was the **largest carnivorous marsupial**. Relentless hunting had driven the wild population to extinction some years earlier.

“ IT IS A GOOD **MORNING EXERCISE** FOR A RESEARCH SCIENTIST TO **DISCARD A PET HYPOTHESIS** EVERY DAY BEFORE BREAKFAST. IT KEEPS HIM YOUNG. ”

Konrad Lorenz, from *On Aggression*, 1966

The last thylacine
Named Benjamin, this was the last thylacine in existence. A predator of kangaroos and wallabies, the species earned an exaggerated reputation for attacking livestock.

American biochemist Wendell Meredith Stanley is the first to **crystallize a virus**, and shows that it is still infectious

November 12 Portuguese neurologist Egas Moniz performs the **first lobotomy**

November 23 Daniel Bovet discovers that **sulfonamide** is the active component of Prontosil

December 12 British Patent Office registers **nuclear chain reaction** for Leó Szilard

February Leó Szilard gives the **atomic bomb patent** to British Admiralty

March Konrad Lorenz and Nikolaas Tinbergen begin collaborating on the study of **animal behavior**

American astronomer Edwin Hubble publishes *The Realm of the Nebulae*, revising his galaxy classification to include **disk galaxies**

June American physicists Seth Neddermeyer and Carl Anderson discover evidence for subatomic particles called **muons**

July 4 Hans Selye recognizes **stress** as a medical condition

September The Palomar Observatory in California begins operating the **Schmidt telescope**

September 7 Benjamin, the last **thylacine**, dies in captivity

November 2 British Broadcasting Corporation launches first regular **high-definition television** service